

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: circular
Species Used	Mouse cells, unspecified		

Before the Pulse

Cell Growth Medium	DMEM	Growth Phase at Harvest	Log phase
		Pre-pulse Incubation	Not given

Wash Solution Phosphate Buffered Saline

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature Room temperature
Electroporation Medium* Not given

Cuvette Gap 0.4 cm

Cell Density 10 (8) cells /250 µl

Voltage 0.40 kV

Volume of Cells 250 µl

Field Strength 1.0 kV/cm

DNA Concentration 20 µg

Capacitor 25 µF

DNA Resuspension Buffer Phosphate Buffered Saline

Resistor (Pulse Controller) Ω none

Volume of DNA 250 µl

Time Constant 1.2 msec

After the Pulse

Outgrowth Medium DMEM

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
 Electroporation efficiencies may be enhanced by use of the Capacitance Extender (provides longer time constants).

Outgrowth Temperature 37 °C

Length of Incubation 48 hours

Selection Method or Assay Used CAT assay

Electroporation Efficiency Not known

Per Cent Survival 20 to 50%

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